
33. Uncertainty

David C. Folch

1. INTRODUCTION

The raw thing that we create is the soul of music and the stories and the feelings of being human. And the technical thing that happens is trying to capture that. And if you try to capture that and you use tools that are less precise, then you get less of it. You know the feeling that you have when you hear something that's really great and it touches your soul? The chances of it really touching your soul are much better if you hear all of it. Neil Young (National Public Radio 2019)

The goal of social science is to explain human behavior using models grounded in observations of people and their environment. However, this empirical effort is constantly being undermined by those being observed. Variation in people and their spatial context makes them difficult to pin down and generalize. In response, we collect more data and build better models so that we can *hear* more of what society has to say. One of the few things we can be certain of is that uncertainties are part of the social world, and by embracing that uncertainty, our explanations have more resonance and clarity, and to that end a greater impact.

Social science mixes the study of complex societal relationships with many of the empirical methods found in the hard sciences. However, social sciences are not hard sciences, and we cannot expect the same level of exactness as our colleagues in the physics and chemistry labs. When a social science regression has an R^2 of 0.95 or 0.99 the researcher should be skeptical of the underlying model specification. The inherent noisiness of social science phenomenon and the lack of complete information means that high model fit often points to issues of autocorrelation, over-fitting or other conceptual problems. Large, precise and representative datasets are the exception not the norm.

Standard statistical diagnostics such as p -values and R^2 provide important indicators of uncertainty in final model results, but largely overlook uncertainty throughout the research process. Data are often themselves the outputs of models, which means that their uncertainty becomes part of downstream results. Therefore, the research starting point is itself uncertain. Even more challenging is that assumptions, conventions and rules-of-thumb used during the research process do not come with error bars. Ignoring all these uncertainties is in effect accepting the potential for unlimited uncertainty in our final results.

This chapter takes an expansive view of uncertainty by asking the question: how confident are you in your results? This simple question touches on the quantitative and qualitative aspects of the research process. It implies striking a balance between statistical tests for variable selection and model specification on one hand, and research questions and social science theory on the other. It considers both measurable error and risk of error when measurements are not possible. Since uncertainty in the spatial social sciences is so broad, this chapter focuses on uncertainty at the beginning and end of the research process. The middle of the process is discussed conceptually, but these matters are covered in detail in Part 2 of this book.

2. UNCERTAINTY FRAMEWORK

The boundless sea of ignorance has shores which we can stand on and map.

Funtowicz and Ravetz (1990)

Uncertainty is a topic of interest across all scientific disciplines, and as such it is characterized in many ways. Vesely and Rasmuson (1984) outline three types of uncertainty in the potentially catastrophic context of nuclear power plant risk analysis. *Parameter uncertainties* include all the vagaries of model inputs ranging from small sample sizes to parameter settings to how experts deal with missing data. *Modeling uncertainties* relate to applicability and precision: whether the model is missing key inputs and if the relationships between inputs are correctly specified; it further includes how the researcher interprets the models. *Completeness uncertainties* occur at the analysis design stage and relate to whether the analyst is aware of the full scope of the problem being studied and the relationships needing to be examined.

In a more general book on science and policy, Funtowicz and Ravetz (1990) present three sorts of uncertainty. *Technical uncertainty* captures the idea of inexactness or variance in an estimate as represented by error bars and significant digits. *Methodological uncertainty* corresponds to unreliability or the amount of confidence in an estimate as measured by a quantitative confidence level (for example, 95 percent) or qualitative indicators such as ‘optimistic.’ *Epistemological uncertainty* relates to our knowledge and the ability of the models we develop to connect to the real world.

Despite Vesely and Rasmuson (1984) and Funtowicz and Ravetz (1990) both addressing qualitative and quantitative assessment in defining uncertainty, the more common views we see of uncertainty are measurable quantities such as bias and variance. Nesselrode and Grimm (2019) explicitly critique the over-quantification of uncertainty in the social and medical sciences. They identify *clarity* and *likelihood* as the two dimensions of uncertainty. Similarly, Fox and Ülkümen (2011) identify *epistemic* and *aleatory* concepts of uncertainty. In both cases the former captures a lack of knowledge of all dimensions of a problem and the latter the probability that a particular outcome will occur. In a model of whether 15-year-olds will eventually graduate college, there is concern about our full understanding of the factors affecting childhood outcomes (clarity), and that two children with the same measured factors can have different outcomes (likelihood). These dimensions of uncertainty are theoretically distinct, but in practice highly related.

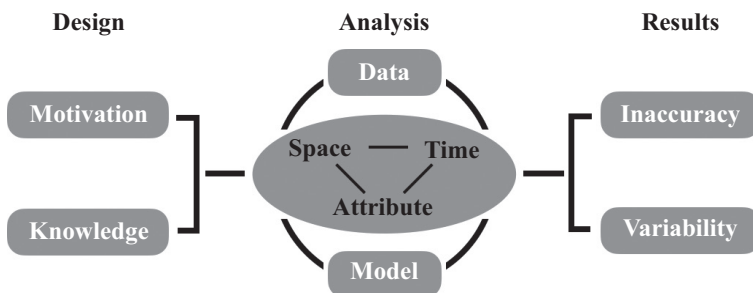


Figure 33.1 Uncertainty framework based on a project workflow

While there is no single way to frame uncertainty, all these characterizations can be viewed through the lens of a research workflow (see Figure 33.1). Some uncertainty issues arise at the project design phase, others at the analysis phase and they finally come together in the results phase. When specifically considering spatial social science, there are spatial, temporal and attribute dimensions that permeate the entire process. This research path framework will help guide the rest of this chapter starting first with space, time and attribute, then moving through the design, analysis and results phases.

3. SPACE. TIME. ATTRIBUTE.

The space, time and attribute dimensions are central to understanding uncertainty in the spatial social sciences. There are direct uncertainties within each dimension, and more complex uncertainties that emerge when the dimensions interact.

3.1 Space

Examples of direct spatial uncertainties abound, and can be grouped into representation, measurement and modeling. Representation covers issues of how real-world objects are encoded by a geographic information system (GIS). Due to storage and measurement limitations, any spatial object in a GIS is an approximation of its real-world counterpart. Cartographic generalization captures a broad suite of techniques for simplifying shapes. These methods might decrease the sinuosity of a coastline or collapse many buildings on a property into a single object. While generally referencing data presentation, generalization has close analogs to the limitations and choices made when measuring the true shape and location of complex objects. Conceptual issues around representation also exist. A house, for example, can be represented as a point or a polygon; a road as a line or polygon. Natural objects, such as a wetland, can have shapes that oscillate with the seasons and evolve over years. All objects can change abruptly after a major natural or anthropogenic event. As objects become more abstract, their representations in a GIS become more unsettled: the idea of a ‘neighborhood’ has long been a core part of social science research (Sharkey and Faber 2014), but neighborhood boundaries appropriate for one study are likely not applicable to another (Weiss et al. 2007; Lehman-Frisch et al. 2012). Therefore, the uncertainty around representation relates to both the limits of computer systems and our fundamental understanding of what objects are (Schneider 1999; Smith and Varzi 2000; Smith and Mark 2003).

Technical issues relate to errors in measuring devices and the depth of knowledge about the represented objects. No matter how clear an object is in the real world, global position system and surveying equipment have accuracy issues that result in incorrect placement on maps (Zandbergen and Barbeau 2011). Bridges over roads and freeway underpasses can be inadvertently encoded as intersections because a planar view of the road network appears to show a connection (Gröger and Plümer 2011). These uncertainties emerge from the raw data entered into the GIS and can be difficult to uncover.

The final category involves modeling space. For example, measuring distance is not straightforward (Miller 2000; Miller and Wentz 2003). As points get further apart, the curvature of the earth impacts the measured distance between them. When distance reflects travel by car, network distance along roads is more relevant than Euclidean distance. There is

also ambiguity when measuring distance between two polygons; should the distance be from centroid to centroid, between outer boundaries or something else (Okabe and Miller 1996)? The representational and technical uncertainties can percolate into the modeling of space. Incorrectly encoded road networks and misplaced points will impact routing algorithms that rely on accurate data for finding paths between locations and defining service territories for warehouses, fire stations and other services (Gaboardi et al. 2019). Hot spot analysis aims to understand the center and spatial extent of phenomenon such as disease and crime; these analyses rely on accurate location data and user defined inputs such as bandwidth (Han et al. 2009).¹

3.2 Time

Direct temporal uncertainty has similar contours to spatial uncertainty but is narrower in scope since time is linear and unidirectional. An event necessarily spans some time, but might be represented as a point in time. Similarly, an event might simply be recorded as happening on some day or in some year but for some research questions, time of day or season within the year is critical. For example, seasonal population fluctuations in a vacation beach town or college town mean that the average population over the year is not representative of any time during the year (Goldschmidt and Dahl 1976). Populations can be quite different in the daytime and nighttime in suburbs and downtowns, which affects service delivery planning (Kobayashi et al. 2011).

3.3 Attribute

Attributes in the social sciences include demographics, business characteristics, government policies, weather, crime, health, tourism and the list goes on. Each of these have direct uncertainties that are domain specific. It is common to hear political pundits say that candidate X is ahead by 2 percent, and then caution that we should not rely too heavily on that finding since the poll has a margin of error of ± 4 percent. Similarly, weather forecasts are always couched as ‘expected’ as opposed to certain.

Uncertainty in political polls and weather forecasts typically relate to data collection challenges such as sampling strategies and measurement errors, these are addressed again later in the chapter. There can also be a more fundamental issue relating to understanding what is being measured. For example, household income can be defined to exclude or include sources such as government assistance or investment earnings. It can also include non-monetary sources such as the value of leisure. The uncertainty in this case does not come from the data themselves, assuming they are well documented and collected, but from misalignment between the research question and a dataset’s definition of income.

In many cases, uncertainty is reported in the attribute dimension, but is caused by spatial or temporal uncertainties. The following section on intersectional uncertainty explores some of these cases.

3.4 Intersectional

Intersectional uncertainty occurs when the space, time and attribute dimensions interact. The most well known of these interactions in spatial social sciences is the modifiable areal unit problem (MAUP) (Openshaw 1977; Fotheringham and Wong 1991). The MAUP occurs when the results of an analysis change when the spatial units used to aggregate the data change. The

two versions of this are scale, that is, dividing a region into more or fewer areas, and zoning, that is, holding the number of areas constant but changing their shapes. Consider a study on racial discrimination in real estate transactions; if discrimination is shown when the city is divided into 1,000 neighborhoods, but not when divided into 500, or discrimination is shown with one configuration of 500 neighborhoods, but not another, then the MAUP is a concern. These issues arise when the boundaries used for analysis are arbitrary, but the underlying spatial processes have true but unknown scales of variation.

The ecological fallacy (Robinson 1950) is another intersectional uncertainty. This is the misconception that statistical relationships between attributes aggregated from individual data hold for the individuals from which the data were aggregated. Demonstrating the problems with correlations, Freedman (1999) uses 1995 data to show that the correlation between percent foreign born and percent high income by US state is 0.52, which might be interpreted as foreign-born residents having higher incomes than native born; however, the correlation at the individual level is -0.05 . The ecological fallacy does not imply that all aggregate variables are problematic, it simply points to their misuse.

Related to the ecological fallacy is the assumption that the aggregate variables for an area are representative of the experiences of all the people living in that area. The neighborhood effects literature is grounded in the idea that the milieu in which a person lives impacts their outcomes (Sampson et al. 2002). A few frameworks directly tackle the challenge of measuring this milieu. Kwan (2012) defines the uncertain geographic context problem as the challenge of not knowing the true geographic context affecting individuals and the uncertainty that thus permeates results. Robertson and Feick (2018) identify the uncertain point observation problem as incorrect context data arising when there is uncertainty about how source-level point data should be grouped. Fowler et al. (2019) define the contextual fallacy, emphasizing that increasing heterogeneity within an area makes contextual inferences less reliable. If the proverbial railroad tracks cut through a geographic unit, then the socioeconomic status on one side of the tracks will be much higher than on the other; this problem is compounded when these dividing lines are not readily observable. Individuals may be living close to one another geographically, but their context, as measured by aggregate data, can be quite a bit different.

As its name implies, the time geography framework (Hägerstrand 1970; Pred 1977) considers the intersection of time and space, or more precisely, the inseparability of the two since all activities occur at some location and at some time. The original framing proposed a space-time cube (two-dimensional space with time as the third dimension) that a person moved through over their day and over their lifetime. This framework helps bound space-time uncertainty: from some given start point in the space-time cube, a person can only reach certain locations in a given time. For example, consider a 14-year-old standing at the front of their school at the end of the school day, there is a finite area that child can reach by foot in 30 minutes. This intersection of space and time uncertainty then ripples into the attribute dimension through similar mechanisms as discussed above in regard to space and attributes.

4. PROJECT: DESIGN

Reports that say that something hasn't happened are always interesting to me, because as we know, there are known knowns; there are things we know we know. We also know there are known unknowns; that is to say we know there are some things we do not know. But there are also unknown unknowns—the ones we don't know we don't know. And if one looks throughout the history of our

country and other free countries, it is the latter category that tend to be the difficult ones. Donald Rumsfeld (Rumsfeld 2002)

At the project design phase there are two key factors at play (see Figure 33.1). *Motivation* relates to having a clear understanding of the project goals. *Knowledge* relates to knowing where uncertainty might arise in the project.

4.1 Motivation

When uncertainty is part of the initial project stages, it can be handled strategically. This starts with the research question and research design, and how these define the eventual project results. A case study approach analyzes a narrowly defined group or location, which allows for detailed study of its many nuances. When done well, the results are highly certain for that case, but uncertainty expands rapidly when applying the findings to situations further and further afield from the specific case. In contrast, a generalized study trades specificity in terms of the subjects being studied for broader applicability. A case study approach might consider socialization skills of students in a particular classroom with fine-tuned outcome measures and detail on each student. A generalized study might use aggregated disciplinary data for all elementary schools in a state. The statewide study can make reliable statewide inferences about general socialization, while the case study can make precise socialization inferences, but only for a narrowly defined subset of the population.

Another dimension of motivation involves allocation of resources (research funding, data sources, expertise and so on). Greater resources make it possible to reduce uncertainty: time and money can be deployed to collect targeted data or increase sample sizes, to access increased compute power or to increase the expertise on the research team. In effect, resources can reduce the uncertainty gap between case studies and generalized studies. With enough resources, socialization skills of individual students in every elementary school in the state could be studied in detail. The relationship between motivation and uncertainty then comes down to a cost-benefit analysis. Ideally the researcher can trade a modest increase (or strategic redeployment) in resources or a modest change in the specificity of the study design for large gains in the certainty of the outcomes.

The design of the American Community Survey (ACS) provides an example of how considering uncertainty, resources and output specificity can influence a change in research design (Spielman et al. 2014). Prior to the ACS, the U.S. Census Bureau collected demographic data on the US population once every ten years as part of the constitutionally mandated decennial census. Every person was asked a short list of questions, while a sample was asked a much longer list of questions. This approach resulted in data (that is, attributes) with little uncertainty, even for small population groups and small geographic areas. The problem was temporal uncertainty when applying the data to non-census years. The ACS ushered in a new era where all households are asked a few questions every ten years, and the sample of people asked the long form questions is spread across the decade. This results in annual data releases: large geographic units (population over 65,000) receive estimates based on one year of data collection, while all places receive estimates based on five years of pooled data. This is a clear win for large places, which now get annual one-year data. For small places there is an explicit trade-off along the temporal dimension: more frequent data releases, but the data now represent a multiyear time window. The temporal grouping of ACS samples illustrates a

strategic decision on the part of the U.S. Census Bureau. They redesigned the decennial census and created the ACS to match customer demand for more frequent data releases, and expected cost savings on the operations side (U.S. Census Bureau 2009).

Figure 33.2 makes the Census Bureau's decision more concrete by illustrating the trade-offs between the three dimensions of space, time and attribute as a continuum of aggregation (or scale). More specific representations of each dimension are near the bottom, with more general ones near the top; and the four thin lines represent the same level of attribute uncertainty for different annual sample sizes. The 'large sample' line near the bottom represents a census, where a large sample simultaneously allows high spatial, temporal and attribute resolution. The 'small sample' line near the top represents the large sacrifice needed in spatial resolution to maintain the desired level of uncertainty. The two 'moderate sample' lines illustrate the trade-off between the three dimensions: a detailed attribute for small areas requires aggregating over time, while a detailed attribute for a narrow time window requires aggregating over space. Although users of ACS data are not able to choose how the U.S. Census Bureau delivers data, the Census Bureau's motivations around uncertainty exemplify the trade-offs researchers face when planning a project.

Figure 33.2 shows that disentangling uncertainty and output specificity is not straightforward. Tobler's First Law of Geography, that 'everything is related to everything else, but near things are more related than distant things' (Tobler 1970, p. 236), contributes to this challenge. On one hand is a reasonable assumption that stronger relationships exist between things in

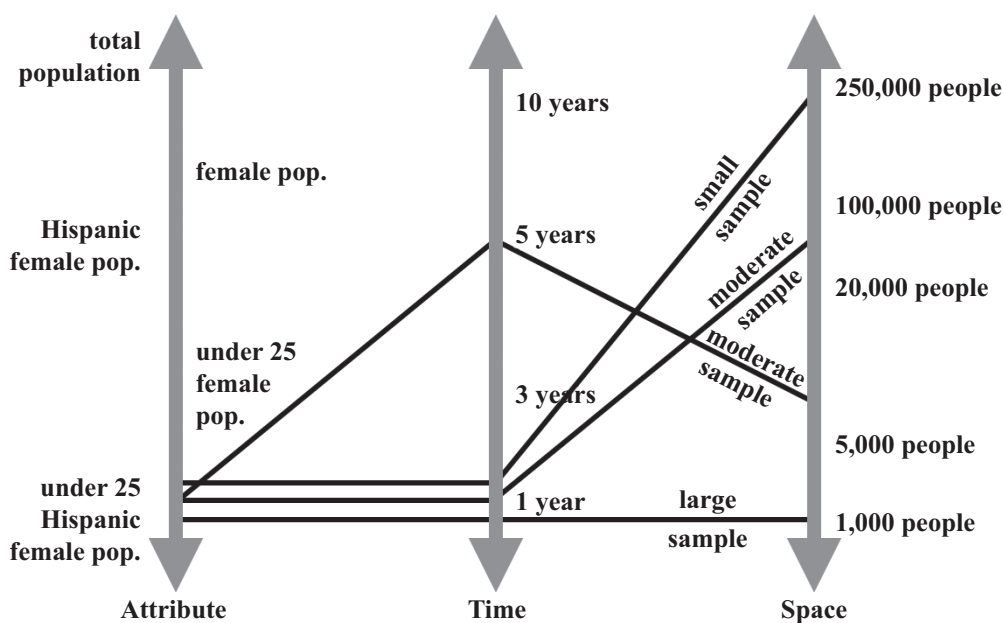


Figure 33.2 *Space, time and attribute uncertainty trade-off in demographic data collection and distribution*

smaller geographic areas than in larger areas. On the other hand, we may only be able to reliably characterize relationships between things in larger areas due to limited data collection resources and locational vagaries (see earlier discussions on time geography and error in data collection). Noisy data stymie analysis on smaller areas, while diluted relationships stymie analysis on larger areas.

4.2 Knowledge

The only uncertainties a researcher can integrate into an analysis at the design phase are the ones they can foresee, that is, the ones in their knowledge base. The ‘unknown unknowns’ that Donald Rumsfeld spoke to are the really challenging ones. Seeking out these uncertainties up front allows the methodology to be designed around them as opposed to retrofitting the analysis later or only being able to acknowledge them, both of which weaken the overall argument. A conscious approach to uncertainty planning makes better use of resources and gives more useful results.

Global position system (GPS) readings are an example of unknown unknowns for many researchers. A GPS is equally accurate on a cloudy or sunny day; in contrast, it is less accurate on the streets of downtown Chicago than it is in an open field (Garnett and Stewart 2015). A person with little knowledge of GPS may send out data collectors to the downtown and suburbs assuming the data will be equally reliable everywhere, or not send out collectors on cloudy days under a misconception of lower accuracy.

Ideally the design phase is both explicit and humble. Uncertainty will be an issue, so the research plan should directly address it. Domain knowledge cannot be understated when building a research plan. The uncertainties that one social science discipline is trained to consider, may be unknown to practitioners of another.

5. PROJECT: ANALYSIS

... In that Empire, the Art of Cartography attained such Perfection that the map of a single Province occupied the entirety of a City, and the map of the Empire, the entirety of a Province. In time, those Unconscionable Maps no longer satisfied, and the Cartographers Guilds struck a Map of the Empire whose size was that of the Empire, and which coincided point for point with it. The following Generations, who were not so fond of the Study of Cartography as their Forebears had been, saw that that vast Map was Useless, and not without some Pitylessness was it, that they delivered it up to the Inclemencies of Sun and Winters. In the Deserts of the West, still today, there are Tattered Ruins of that Map, inhabited by Animals and Beggars; in all the Land there is no other Relic of the Disciplines of Geography. Jorge Luis Borges (1998)

The transition from project design to analysis implies operationalizing the earlier planning decisions through specific data and models. At a base level, facts about the world are being collected, which in principle are certain. The challenge is that most of these root facts are mediated by an algorithm or an instrument grounded in a model that can introduce uncertainty. Figure 33.3 highlights the complex relationship between data and models. It presents a four-step process, which starts with planning, moves to collection, then processing and finally distribution. This process applies to both primary data collection and secondary data analysis. If viewed as an eight-step process, it allows hybrid projects to start at any step.

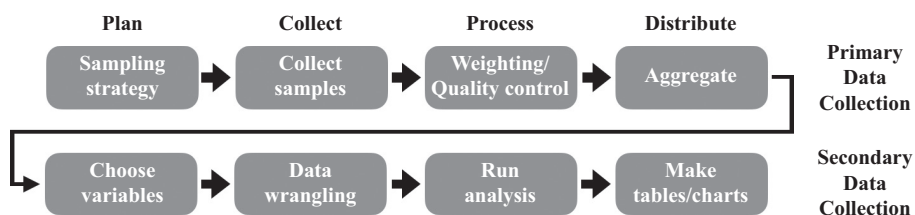


Figure 33.3 *Data and model process*

5.1 Primary Data Collection

The primary data collection track in Figure 33.3 starts with designing a sampling strategy and collection protocols. This step dovetails with the project design phase discussed earlier. The next step is data collection. Challenges here are more mechanical: transcription typos, instrument calibration or malfunction, and so on. The processing step accounts for issues or weaknesses resulting from the previous two steps. Processing actions can be split into systematic and irregular. One systematic approach is weighting, which applies more or less weight to account for how prevalent an observation is in the population. Quality control identifies irregular observations such as outliers. When done well, these actions can reduce uncertainty. The final step compiles the information for distribution. This could be done for many reasons including building convenient datasets for final users or to intentionally obscure detail. Aggregating data and noise infusion are approaches that increase uncertainty with the goal of protecting respondent privacy.

Ideally, the results of primary data collection are fit for their intended purpose. This implies minimizing the uncertainties around that purpose, and possibly (and reasonably) ignoring uncertainties that might result if the data were used for other work. Researchers in the secondary data track rely on primary data providers to document their processes and purposes. Even a well-designed primary data collection process can result in a secondary dataset that introduces considerable uncertainty due to the misalignment of purposes.

5.2 Secondary Data Usage

The secondary track begins with variable selection and model planning, and again dovetails with the project design phase. Ideally, the sources of secondary data are scrutinized and the pros and cons weighed. At this stage, the goals of the project need to be balanced against the appropriateness of the data. This moves into what some have called ‘data wrangling.’ Like the primary data track, this is a more mechanical step of collecting, managing, organizing and prepping the data for the next step; it often accounts for most of time spent on a project (Wickham 2014).

The next step is to run the actual analysis, which might be a regression or machine learning model. It should answer the research question(s) while also accounting for the uncertainties accumulated along the way. The literature on formal model specifications and diagnostics is well developed and expanding. The final step involves summarizing the detailed results: the creation of charts, tables, dashboards and the like open the results to wider audiences.

Any individual project could start at any of the eight steps presented in Figure 33.3. Historical social science projects often start at the third step by digitizing historic maps, raw census documents or other archival material (for example, Logan et al. 2011). Since users of secondary data have no control over how data is collected and processed by a government statistical agency (for example, Spielman et al. 2014), a hospital (for example, Hagger-Johnson et al. 2015), a property tax assessor (for example, Krause and Bitter 2012), a social media company (for example, Arribas-Bel 2014) and so on, it may be tempting to ignore the details preceding the researcher's active role. However, decision makers relying on secondary data track results expect that the findings account for the uncertainties from all eight steps.

6. PROJECT: RESULTS

To this point the chapter has been about the manifold causes of uncertainty and being proactive about research decisions. Although results were discussed previously, in this section we cover the two broad manifestations of uncertainty: inaccuracy and variability.

6.1 Inaccuracy and Variability

Inaccuracy relates to a result not matching the true value. Variability relates to dissimilarity between repeated results. Different disciplines use these terms differently, with inaccuracy often associated with bias and variability with precision. In some cases, (in)accuracy represents the combination of bias and precision, while uncertainty only refers to variability. This chapter defines uncertainty in results broadly to include all situations where it is reasonable to be uncertain about research findings.

Figure 33.4.a1 shows the case of low inaccuracy and low variability with a spatial point pattern packed around the true location (marked with an X). Consider setting a GPS on a park bench and repeatedly recording the latitude/longitude pair it gives. If the average of all the readings is close to the actual location of the park bench, then the GPS has low inaccuracy (that is, high accuracy). If the readings are packed together, the GPS has low variability. However, inaccuracy and variability are independent. The readings could be packed together, but all located far from the bench; this would imply low variability and high inaccuracy. Alternatively, the readings could be scattered all around the park, but when averaged are right on the park bench; this indicates high variability and low inaccuracy. Of course, the worst case is readings scattered far from the bench giving high variability and high inaccuracy. The four b subfigures in Figure 33.4 represent the same concepts, but in terms of distributions. The further the distribution is from the true value (red line) the more inaccurate. See color figure in e-chapter for more clarity. The wider the distribution, the higher the variability. Consider an example of a person wearing a GPS while running multiple laps around a 400 meter track. If the x-axis of the b subfigures is meters and the red line positioned at 400 meters, Figure 33.4.b3 would indicate a GPS that reports similar overestimates of the distance for each lap.

6.2 Measuring Uncertainty

An ideal primary data collection project incorporates 'ground truth' to calibrate the data collection procedure. In this process, collected data are compared to known true values to measure

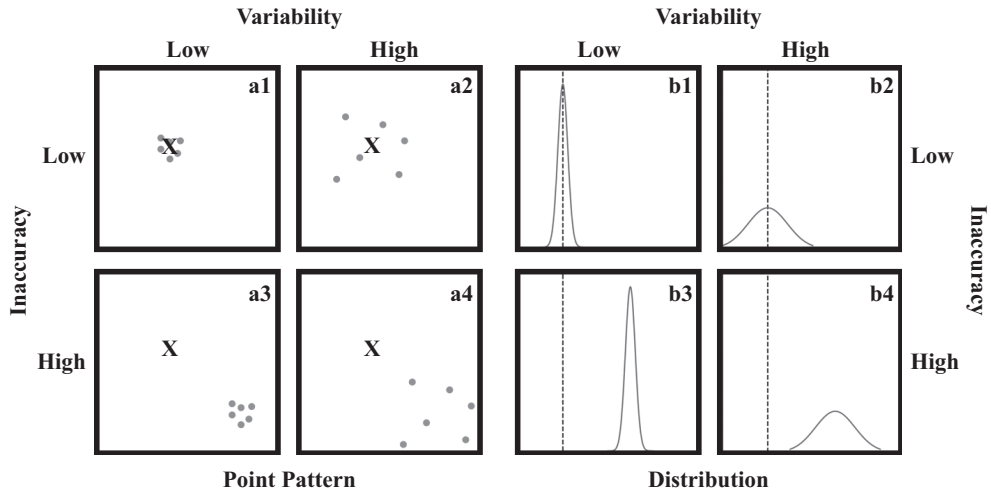


Figure 33.4 *Inaccuracy and variability*

inaccuracy and variability. When applied to model output (lower track of Figure 33.3), the ‘collected data’ are the modeled values and the true values are the corresponding input data.

Root mean squared error (*RMSE*) is a common measure that gives a single value for the combined impact of inaccuracy and variability. Furthermore, data quality and model quality research both apply it using the following formulation:

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}}.$$

RMSE computed for primary data collection defines $\hat{y}_i$ as a value collected at location i (or for person i , business i and so on) and y_i is the ‘true’ value for i collected by some other means. For example, $\hat{y}_i$ could be the distance measured using a GPS device, and y_i the distance from a higher accuracy source (see ASPRS 2015 for more examples of spatial accuracy measurement). When *RMSE* is computed for a model, $\hat{y}_i$ is the predicted value for i from the model and y_i is the observed value for i . In a regression framework, $\hat{y}_i - y_i$ is the residual for observation i . In a cross-validation framework, y_i could come from either the training or testing data.

Variability can generally be measured reliably with a statistic like standard deviation. In contrast, measuring inaccuracy is difficult since the true value is rarely known. In Figure 33.4 we generally do not know where the black X or dotted line is. In practice, we estimate that location with the data available, meaning that a model that fits the data well may be unreliable when presented with new data. A large random sample typically gives low variability results that accurately describe a population. However, many research budgets do not allow for either large or random sampling. Strategies such as stratified sampling and weighting described earlier can leverage theory to help reduce uncertainty in the input data and model results. Returning to the planning phase of the project, having a solid understanding of the project goals helps predict how uncertainty will manifest at the end of the project.

7. LIVING WITH UNCERTAINTY

Give them the third best to go on with; the second best comes too late, the best never comes.

Sir Robert Alexander Watson-Watt

To describe why the British should deploy his suboptimal radar system in the late 1930s to defend against enemy aircraft, Sir Robert Alexander Watson-Watt invoked what he called the ‘cult of the imperfect.’ He knew Great Britain needed a workable early warning system immediately, and unending tinkering to get slightly better results could end with no Great Britain left to defend (Williams 2017). The transparency implied by his statement applies to the inevitable uncertainty that will be present in any spatial social science project, even after reasonable mitigation attempts have been made.

7.1 Sensitivity Analysis

One of the most common approaches for understanding the impact of uncertainty is sensitivity analysis (Chattoe et al. 2000). Sensitivity analysis is a technique that involves changing model inputs to understand the effect on results. This is a particularly appealing strategy as models become more complex and interactions between data and model parameters are difficult to predict from theory. The one-at-a-time strategy changes one input and then reruns the model. If small perturbations in an input cause large changes in the result, then the quality of this input warrants particular attention. Again, we return to the idea of cost-benefit analysis; allocating limited research resources to the inputs that are most influential on the result. More complex models may require changing multiple inputs at the same time to capture interactions in the changes.

The sensitivity analysis approach is not limited to the choices made by the researcher. If the input data come with margins of error or other uncertainty information, this can be used to create variation in the sensitivity analysis. This calls back the aphorism, ‘garbage in, garbage out’: if the underlying data, which are out of the researcher’s control, are unreliable, then the overall results are unreliable.

7.2 Transparency

Reproducibility underlies the sensitivity analysis framework. Reproducibility means that the researcher can reproduce their results using their own code and data. If the workflow is not easily reproducible, it is not possible to conduct a robust sensitivity analysis. A related concept is replicability, which is the robust and rigorous documentation of the workflow so that an independent researcher can arrive at the same results (Broman et al. 2017). In the social sciences, and especially spatial social sciences, both of these efforts are complicated when the study area changes. Similar results across locations reduce uncertainty in the initial findings, but dissimilar results may simply indicate a change in context. No matter how reproducibility and replicability are being employed, they add transparency to the workflow by allowing the research to be validated or extended (see Chapter 31).

Qualitative descriptions of the data and models can also increase transparency. Researchers often have a broad understanding of the uncertainty in their data, especially when they do the data collection. Limited resources and other challenges often mean that data are collected under suboptimal conditions. Clear descriptions and estimations on the direction and magnitude of inaccuracy and variability in the data can help the audience understand how broadly to apply the results. These qualitative descriptions give future researchers an understanding of the gaps needing to be filled to broaden the applicability of the work. If the data collection process under-sampled lower income households, one side of the city, renters and so on, these facts can guide both policymakers and future researchers.

Uncertainty can also be reduced by using ranges instead of point estimates. There is less uncertainty in predicting that a stock will end the day between \$103 and \$107, as opposed to exactly \$104.35. Demographic data providers also use this technique when they estimate counts of households in income ranges, for example, . . . , \$20 000–29 999, \$30 000–44 999, The wider the ranges, the more confidence in the estimated counts; it also helps protect privacy of the surveyed respondents. A model might reliably work only over certain ranges of the data. In a model predicting house prices, it may be enough to reliably predict if the house will have a high, medium or low price, but the uncertainty within categories does not have a substantive impact on decisions relying on that model. This approach comes with costs since wider ranges around predictions and estimates make them less useful for decision makers.

The challenge of representativeness is an important criticism made of nontraditional datasets coming from social media and other newer data sources. They often contain a volume of data that rivals (or exceeds) that collected by government statistical agencies, which can give a false sense of representativeness. However, these data are not collected with any type of sampling strategy as they are simply the contributions of that platform's users (Arribas-Bel 2014). As previously stated, many datasets are not universally representative, especially bespoke samples collected for a particular research project (Faugier and Sargeant 1997); therefore, the newer sources have characteristics in common with some traditional data. Further complicating matters is that these new data sources are noisier than what comes from researchers and statistical agencies, thus relying on the researcher to screen the data before use; in effect starting at the third step in Figure 33.3 without having control of the first two steps. The challenge and opportunity for the researcher is then to identify if these data are representative of some experience or subset of the population or if it can be combined with other data to give a more robust characterization of the population.

8. CONCLUSION

This chapter began by recognizing that the human condition is messy, and that uncertainty is part of any spatial social science research effort. This does not absolve researchers from considering uncertainty in their work, but it does free them from some unattainable uncertainty free result (Couclelis 2003). Ideally, researchers implement state of the art methods using the best available data to produce work that balances uncertainty and timely communication of new ideas. Overarching all of this is the communication of uncertainty through transparency.

NOTE

1. Sinton (1978) provides early foundations on spatial uncertainties and the interrelationship between space, time and attribute and van Oort (2006) provides a summary of specific manifestations of spatial uncertainties.

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